

Exploiting different electricity markets via highly rate-mismatched modular electrochemical synthesis

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Mitigating the inherent spatio-temporal stochasticity and intermittency of renewable power is key for enabling the decarbonization of the power grid and motivates the development of flexible technologies that can shift power demand and supply across space–time and scales. Here we develop an electrochemical synthesis strategy capable of providing demand (load) flexibility at different timescales by participating in multiple electricity markets (day ahead, real time and frequency regulation). Using a fast proton-conducting redox material, copper hexacyanoferrate, highly rate-mismatched modular electrochemical synthesis was achieved by decoupling half reactions with different intrinsic kinetics to produce chemicals under drastically different reaction rates and timescales: the fast hydrogen evolution reaction and slow persulfate production reaction. Such a strategy enables flexible participation in different electricity markets and can reduce electricity cost of chemical production by 30–40%. These results open a conceptual strategy for flexibly integrating modular electrochemical manufacturing processes into the fluctuating power grid to achieve more economical and sustainable operations.

Decarbonizing the power grid requires the widespread adoption of renewable (wind/solar) power sources^{1,2}. Mitigating the unpredictable and intermittent nature of renewable power presents a grand challenge, driving the unprecedented restructuring of the power grid. The multi-scale and spatio-temporal behaviours from this restructuring will require the power grid to harness flexibility in diverse forms (demand, supply, storage) and enable a reliable and sustainable power supply to society^{3,4}. Lack of power grid flexibility is currently manifesting as large amounts of renewable power that remain stranded (cannot be absorbed by the grid) or even the curtailment of renewable electricity generation^{5,6}; stranded power is projected to grow as more renewable power is injected into the grid and hinders the ability to decarbonize the grid⁴. Independent system operators operate different electricity markets to harness and remunerate flexibility provided at different times and network locations⁷. For instance, in

markets such as the day-ahead market (DAM) and real time market (RTM), electricity prices are updated every hour or 5–15 minutes to promote balancing of power supply and demand (Fig. 1a). The frequency regulation (FR) market is a critical ancillary service market that operates at a 2 s time resolution and is designed to secure balancing at fast timescales⁷. The need for FR services continues to increase as more renewable power is absorbed in the grid. The price signals from these markets, particularly negative prices resulting from stranded power, indicate when and where resources are most valuable and provide an economic incentive for technologies that can use/supply power⁸. Therefore, flexible technologies that can participate in markets at different times/locations can help the grid absorb more wind/solar power and economically monetize this service⁹. For instance, manufacturing facilities that can adjust power demands (loads) in real time can provide this service.

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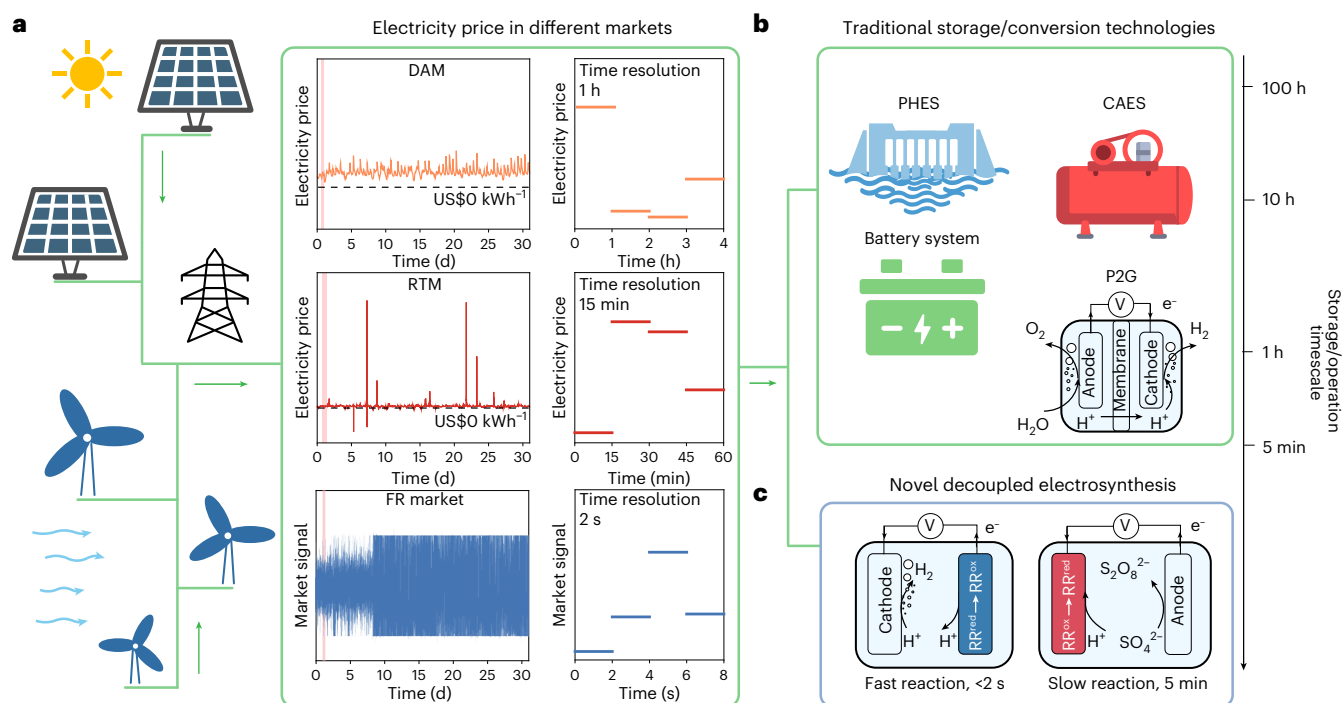


Fig. 1 | Participation in dynamic electricity markets using different technologies. **a**, Different electricity markets are used by independent system operators to balance the power grid and mitigate the risk that results from the variability of solar and wind power at different timescales. For example, the DAM with an hour resolution and RTM with a finer time resolution (5 or 15 min), are used to (re)balance supply/demand by correcting plans. The FR market with a time resolution of 2 s aims to secure balance at fast timescales. The price signals of these markets indicate when and where resources are most valuable; technologies capable of responding to such signals can benefit economically. **b**, Flexible technologies with different storage/operation timescale, such as

PHES, CAES, battery systems, P2G and decoupled electrochemical synthesis, could participate in multiple markets and at different times/locations to help the grid absorb wind/solar power and monetize this service. Power to hydrogen using proton exchange membrane electrolyzers is shown as an example of P2G. **c**, The decoupled electrochemical synthesis could effectively participate in the FR market, which is not possible for traditional systems with long storage/duration time. Here, RR^{ox} is the oxidized RR electrode and RR^{red} is the reduced RR electrode. The green arrows show the electricity flow from generation to demand. The green or blue box shows the traditional storage/conversion technologies or novel decoupled electrochemical synthesis to handle power fluctuations.

Various technologies have been developed to handle power fluctuations in the grid, such as hydroelectric storage (PHES)^{10,11}, compressed air energy storage (CAES)^{12,13}, battery systems^{14,15} and power to gas (P2G)^{16,17} (Fig. 1b). These technologies could strategically convert excess power from the grid (or power at low or negative prices) into potential or chemical energies and then use stored energies during power scarcity (or power at high prices). Compared with PHES or CAES with long storage/operation timescale, electrochemical technologies with a short storage/operation timescale, especially P2G, can lead to spatially distributed power demands and to sources of demand flexibility. Recent results show that incorporating proton exchange membrane electrolysis capable of adjusting power demands in response to price signals in the DAM and RTM to produce hydrogen in methanol synthesis processes can lower the hydrogen cost production by 31% from that via the conventional steam-methane reforming process¹⁸. However, a fundamental issue of such an electrochemical system with coupled half reactions is that the slow half-reaction constrains the system response; from a power grid perspective, this can limit the system's ability to respond to market signals that are faster than the limiting time constant. The mismatch between the timescales of paired half reactions can also lead to system inefficiencies (parasitic reactions)^{19,20}. Instead of the coupled electrochemical system, decoupling paired reactions using redox-active materials^{21–26} or redox reservoirs (RRs)^{27–29} allows for flexible pairing of different half reactions potentially to participate in different electricity markets. In principle, using such a strategy, half reactions can be performed at desired rates and under various timescales to provide more demand flexibility for participation in different electricity markets and process modularization of chemical supply chains.

Here we present a highly rate-mismatched modular electrochemical synthesis (ModES) process that can effectively participate in DAM, RTM and FR markets (Fig. 1c) and reduce electricity costs. We decouple the conventionally tightly coupled hydrogen evolution reaction (HER) with fast kinetics and persulfate production reaction (PSR) with slow kinetics and use a RR electrode with fast kinetics to separately pair with these two half reactions. The electricity in spikes of a few seconds at a low or even negative price can be used to produce H₂ and charge the RR electrode quickly. The electrons/ions stored in the RR electrode enable slower persulfate (S₂O₈²⁻) production over a few minutes. A proton battery material that shows fast proton transport and storage via the Grotthuss mechanism, copper hexacyanoferrate (CuHCF), is demonstrated as the RR electrode with excellent rate capability and cycling stability even in mismatched charging/discharging conditions. The developed computational modelling framework shows that this highly rate-mismatched ModES system operating under a flexible power demand could reduce electricity costs by 30–40% compared with the baseline and even generate revenues by participating in the FR market. These results confirm the promise of modular electrochemical synthesis for providing flexibility to dynamic electricity markets and show how flexible electrochemical systems can open revenue opportunities.

Design of the highly rate-mismatched ModES system

We design a highly rate-mismatched ModES system to produce value-added chemicals under different timescales to participate in dynamic electricity markets. The key to such a system is the use of solid-state redox materials that can achieve fast storage/release of

electrons and ions and decouple electrochemical half reactions with different intrinsic kinetics. As the initial demonstration, we choose HER, one of the fastest electrochemical reactions that produces green hydrogen fuel, as the model reaction to participate in the FR market, and PSR, a widely used electrosynthesis with slow intrinsic kinetics (estimated market size of US\$750 million in 2027)³⁰, as the other model reaction to participate in DAM and RTM. Compared with conventional persulfate production (Supplementary Fig. 1), here flexible HER and PSR with different reaction rates are achieved in a membrane-free, decoupled fashion mediated by a RR electrode (Fig. 1c): proton-consuming HER pairs with the oxidation of RR that releases protons at a fast reaction rate. Then the reduction of RR that uptakes protons pairs with the slower PSR in the other electrochemical cell. The RR electrode can transport protons from the PSR cell ($\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$) to the HER cell (Cell_{H_2}) to maintain proton balance. Because dynamic electricity markets exhibit price fluctuations at different timescales across seconds to hours, stable RRs with fast kinetics and selective proton transport must be developed for highly rate-mismatched electrochemical productions.

Electrochemical performance of the proton battery material

The rate capability of energy storage materials is strongly influenced by the charge carriers. Compared with metal ions, protons have a smaller mass and ionic radius and exhibit fast diffusion kinetics via the unique Grotthuss mechanism (Fig. 2a)^{31,32}, where protons jump from one water molecule to the next through the hydrogen-bonding network³³. Unlike homogeneous redox materials that are challenging to be charged/discharged within seconds, heterogeneous redox materials with defects^{34,35} or large tunnels³⁶ may form inner hydrogen-bonding networks and achieve fast Grotthuss conduction inside solids. Therefore, considering the electrochemical activity and stability of redox materials in acidic solutions, we choose Prussian Blue analogue copper hexacyanoferrate (CuHCF) that could form a crystal water network inside the framework structure³⁷, which showed promise for fast redox kinetics³⁸.

Nanostructured CuHCF material (Fig. 2b) was synthesized via a co-precipitation method (Methods). The identifiable powder X-ray diffraction peaks, (200), (220) and (400), closely aligned with the standard pattern of the cubic PBA phase (JCPDS number 52-1907) (Supplementary Fig. 2). The chemical compositions of the as-synthesized CuHCF material, $\text{K}_{0.06}\text{Cu}_{1.5}[\text{Fe}(\text{CN})_6] \cdot 5.5\text{H}_2\text{O}$, was revealed by thermogravimetry analysis (analysing water content; Supplementary Fig. 3) and inductively coupled plasma-optical emission spectroscopy (analysing element compositions; Supplementary Table 1).

We then studied the electrochemical behaviours of the CuHCF material in 2 M H_2SO_4 solution. Cyclic voltammograms (CVs) showed four pairs of reversible redox peaks, where OI/RI was attributed to $\text{Cu}^{\text{II}}/\text{Cu}^{\text{I}}$, and other peaks were from $\text{Fe}^{\text{III}}/\text{Fe}^{\text{II}}$ (Fig. 2c)^{39,40}. The peak current in CVs, I , was analysed based on equation $I = av^b$, where a was the coefficient and v was the scan rate. The b values for the anodic peaks (Fig. 2d) and cathodic peaks (Supplementary Fig. 4) were close to unity, indicating strong capacitive behaviours with fast kinetics^{41,42}. The CuHCF electrodes exhibited excellent capacity retention of 75.3% at even 1,500 C rate (a short (dis)charge period <2 s) (Fig. 2e and Supplementary Table 2). Note that the n C rate means that the current when the (dis)charge period of the electrode is $1/n$ h, and the 1 C rate is defined as 65 mA g^{-1} based on the theoretical capacity of CuHCF (65 mAh g^{-1}). The linearly increasing potential hysteresis of CuHCF with increasing current densities (Supplementary Fig. 5) revealed that its rate capability was limited by uncompensated resistance in the electrochemical cell rather than the kinetics of proton storage/release. Galvanostatic cycling tests confirmed the long-term stability of CuHCF at 1 A g^{-1} (15 C) with 0.006% decay per cycle (Supplementary Fig. 6a) and at 5 A g^{-1} (75 C) with 0.005% decay per cycle (Supplementary Fig. 6b).

We further evaluated the capacity and Coulombic efficiency (CE) of CuHCF at mismatched charging/discharging rates. No matter how fast the charging rate was, CuHCF showed CE above 90% when the discharging rate was above 1.5 C (Fig. 2f and Supplementary Fig. 7). Reduced CE might result from the self discharge of the electrode at a high charging rate (Supplementary Fig. 8). The CuHCF electrode also showed capacity retention above 78% (compared with the capacity at 1 C rate) and good cycling stability at mismatched rates if the ratio of the charging rate/discharging rate stayed below 500 (Fig. 2g and Supplementary Fig. 9). These results suggest that rate-mismatched production of persulfate and hydrogen with a time ratio from 1:1 to 100:1 should be achievable with a high CE and capacity retention using the CuHCF electrode (Fig. 2h). Specifically, when the time ratio of PSR to HER is 100:1 with an HER rate of 1,000 C and a PSR rate of 10 C, the HER reaction would finish in 2 s to enable its participation in FR market to receive flexibility incentives. The excellent long-term stability and fast proton transport suggest the suitability of CuHCF for rate-mismatched electrochemical productions.

Investigation of HER and PSR in conventional electrolysis

We separately investigated HER and PSR in 2 M H_2SO_4 solution considering their fast kinetics and high efficiency in acidic solutions⁴³. Pt electrode, widely studied for acidic HER⁴⁴, was chosen as the catalytic electrode for H_2 production. The linear sweep voltammetry (LSV) curve of the Pt electrode shows an overpotential as low as 43 mV to achieve the current density of 10 mA cm^{-2} (Supplementary Fig. 10a). Chronoamperometry curves at different applied potentials show nearly constant currents without noticeable decay (Supplementary Fig. 10b), suggesting excellent catalytic stability.

To minimize the competitive oxygen evolution reaction (OER) and possible chemical corrosion^{43,45,46}, we chose a stable boron-doped diamond (BDD) electrode with a high overpotential for OER and a good activity for PSR (details on electrode fabrication in Methods and Supplementary Fig. 11). The LSV curve of the BDD electrode shows an increased anodic current density with increasing applied potential and a high current density of 85 mA cm^{-2} at 3.10 V vs standard hydrogen electrode (SHE) (Supplementary Fig. 10c). As the applied potential increases, the Faradaic efficiency of PSR ($\text{FE}(\text{S}_2\text{O}_8^{2-})$) increases and reaches above 80% when the applied potential is above 2.84 V vs SHE (Supplementary Figs. 10c,d and 12). Moreover, stable bulk PSR electrolysis over the BDD electrode can accumulate $\text{S}_2\text{O}_8^{2-}$ (Supplementary Fig. 10d), confirming the excellent activity and stability of the BDD electrode.

Highly rate-mismatched ModES production

We then demonstrated a prototype rate-mismatched ModES production of H_2 and $\text{S}_2\text{O}_8^{2-}$ mediated using a CuHCF RR electrode in two electrochemical cells (Fig. 3a; Methods). During rate-mismatched ModES cycles (Fig. 3b and Supplementary Fig. 13), in Cell_{H_2} , HER at the Pt electrode accompanies the oxidation of RR that releases protons; then in $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$, PSR at the BDD electrode accompanies the reduction of RR that uptakes protons. Here we assume the FE of HER to be 100% since the only possible product during the cathodic reaction is H_2 . The FE ($\text{S}_2\text{O}_8^{2-}$) was calculated based on the measured concentration of the produced persulfate in $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$. During ModES cycles, the average potential difference between the cathode and anode in each cell determines the average cell voltages (V_{cell}) (ref. 29). We use the voltage efficiency (VE) to describe the energy efficiency of the ModES^{27,29}:

$$\text{VE}(\%) = V_{\text{DIR}}/V_{\text{ModES}} \times 100\% = (V_{\text{S}_2\text{O}_8^{2-}} + V_{\text{H}_2} - \Delta V) / (V_{\text{S}_2\text{O}_8^{2-}} + V_{\text{H}_2}) \times 100\% \quad (1)$$

where V_{DIR} is the potential difference between the HER at the Pt electrode and PSR at the BDD electrode, respectively; ΔV represents the average potential difference of the RR electrode in Cell_{H_2} and $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$.

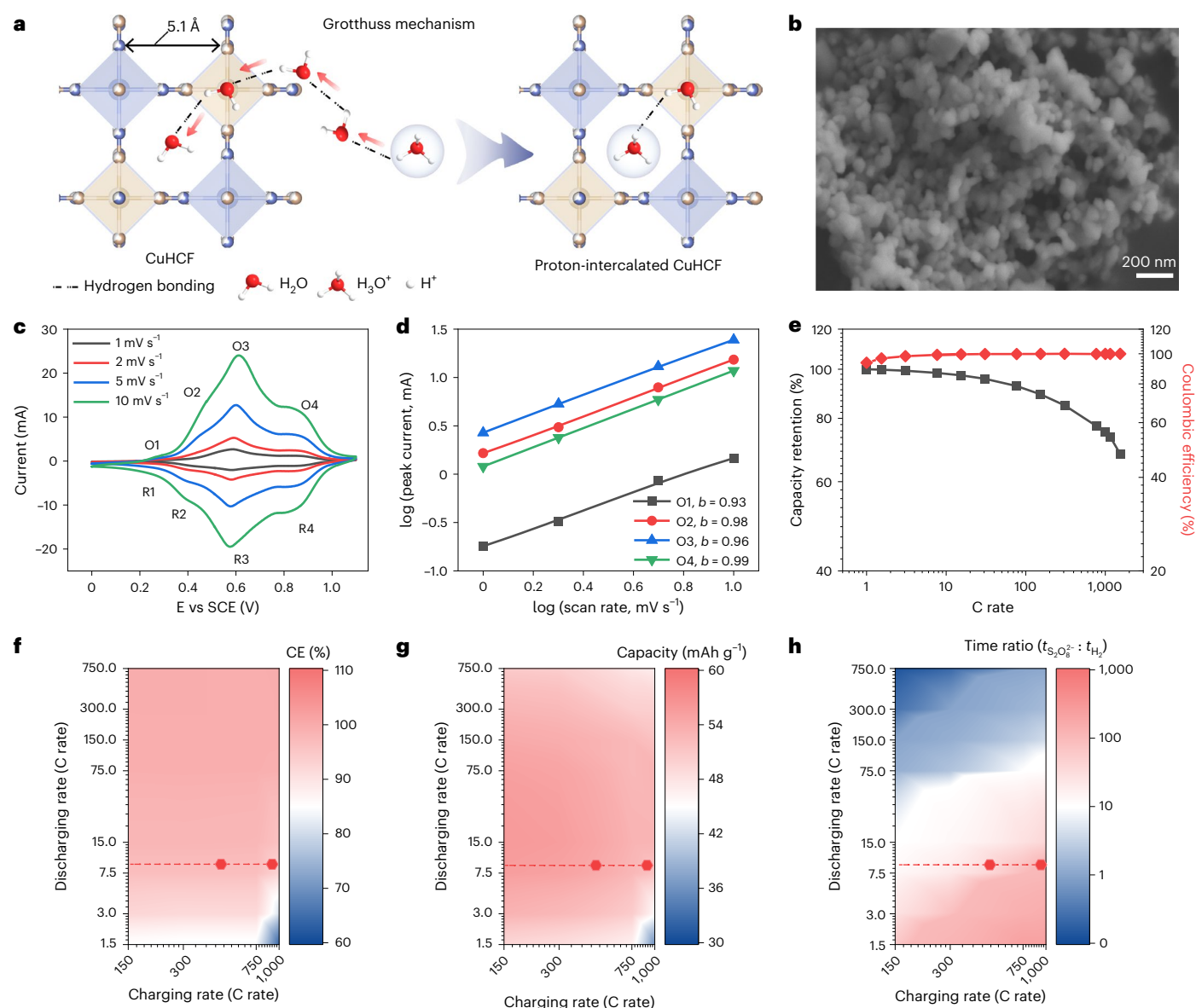


Fig. 2 | Structural and electrochemical characterizations of CuHCF as a fast proton battery material. **a**, Schematic illustration of the Grotthuss mechanism in which protons transport through the hydrogen bond network. The light blue polyhedral is Cu-centred, and the light brown one is Fe-centred. The blue atom is N, and the brown atom is C. **b**, Scanning electron microscopy (SEM) image of as-synthesized CuHCF nanoparticles. **c**, Cyclic voltammograms of the CuHCF electrode in 2 M H₂SO₄ solution at 1 mV s⁻¹, 2 mV s⁻¹, 5 mV s⁻¹ and 10 mV s⁻¹. O1, O2, O3 and O4 label different oxidation peaks, and R1 to R4 show corresponding reduction peaks. **d**, Analysis of the anodic peak current (*I*) as a function of the

scan rate (*v*) in logarithmic scale to extract the exponent *b* values according to the equation $I = av^b$. **e**, Rate capability and corresponding Coulombic efficiency of the CuHCF electrode in 2 M H₂SO₄ solution. **f, g**, Coulombic efficiencies (**f**) and capacities of the CuHCF electrode (**g**) at different charging/discharging rates (each heat map was interpolated from 32 data points). **h**, Estimated time ratios of the persulfate production to hydrogen production using the CuHCF electrode at mismatched rates (the heat map was interpolated from 32 data points). Red dashed lines in panels **f–h** represent targeted mismatched ModES processes with a fixed discharging (PSR) rate of 10 C.

We started ModES production of H₂ and S₂O₈²⁻ from equal rates of HER/PSR and finally achieved a high mismatched production rate (100:1) of HER/PSR: the PSR rate was kept at 10 C based on the capacity of the CuHCF RR electrode, and the HER rate increased from 10 C to 1,000 C (Supplementary Figs. 14–17 and Supplementary Table 3). During the first five cycles with equal rates, the average RR potentials (*E*_{RR}) in both cells were nearly 1.0 V vs SHE (Fig. 3c), and the Δ*V* was only 20 mV (Supplementary Fig. 14), confirming the fast proton kinetics of CuHCF. The corresponding *V*_{cell} showed repeatable values of 1.00 V and 1.91 V with fluctuations less than 10 mV in Cell_{H₂} and Cell_{S₂O₈²⁻}. Moreover, an average CE of RR of 99.4%, VE of 99.7% and FE (S₂O₈²⁻) of ~75% (Fig. 3d) indicate the high energy efficiency of the ModES system. During cycles

6 to 10 when the HER rate increased to 100 C, *E*_{RR} and *V*_{cell} in Cell_{H₂} increased by about 20 mV and 90 mV (Fig. 4c and Supplementary Fig. 15), respectively, which could result from the RR electrode's iR drop and overpotential for HER. The *E*_{RR} and *V*_{cell} in Cell_{S₂O₈²⁻} showed nearly no change. The average CE, VE of the system and FE (S₂O₈²⁻) were like those seen in cycles 1–5, confirming that RR enabled a stable decoupled operation at a 10:1 ratio of the HER rate/PSR rate.

We then performed highly rate-mismatched ModES production at the HER rates of 500 C and 1,000 C and the PSR rate of 10 C, using a smaller CuHCF RR electrode. When the ratios of mismatched rates were 50:1 and 100:1, fast HER can be achieved in 4.77 s (Fig. 3b and Supplementary Fig. 16) and 1.87 s (Fig. 3b and Supplementary Fig. 17),

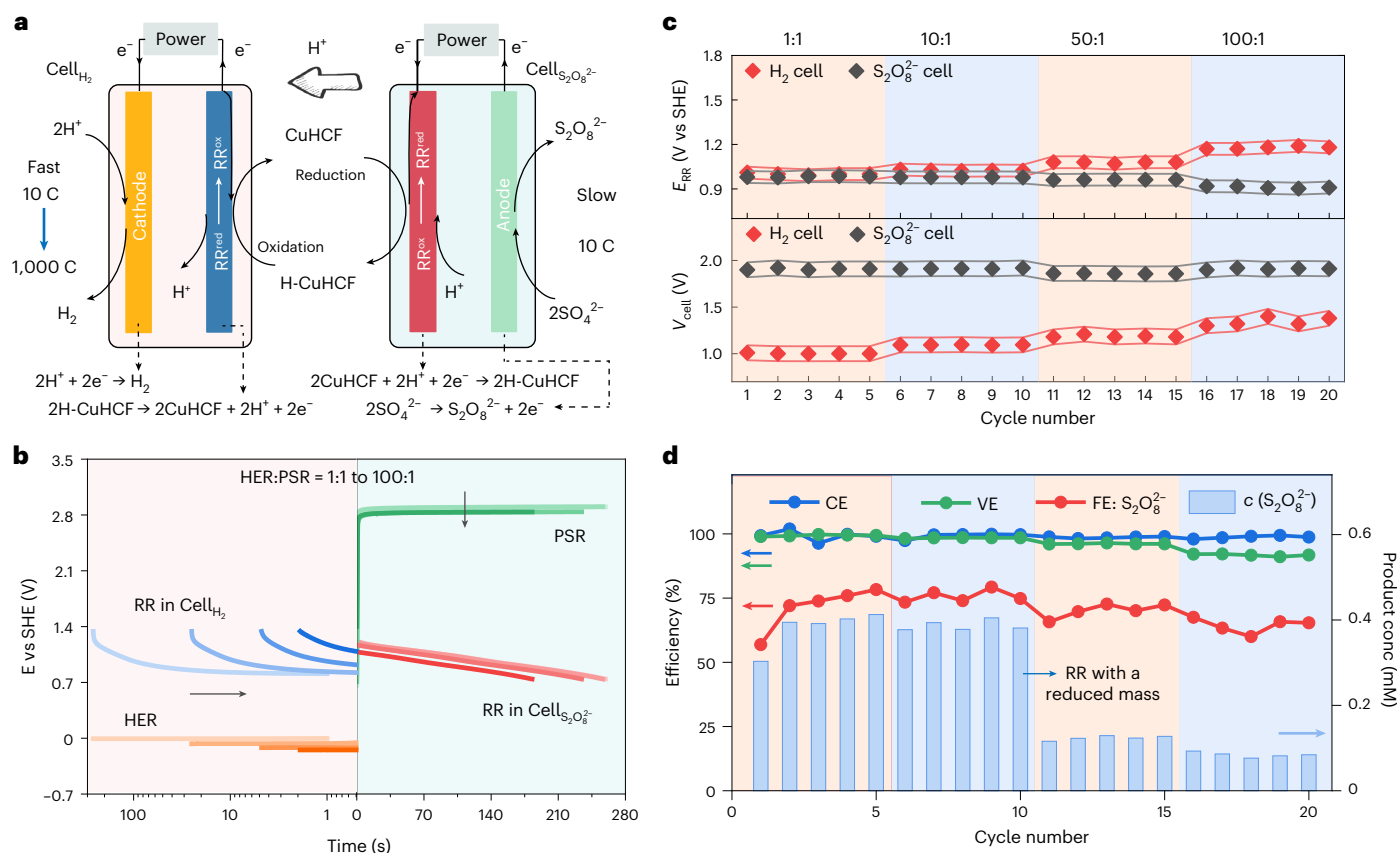


Fig. 3 | ModES with highly mismatched reaction rates. **a**, Schematic illustration of the ModES system with highly mismatched reaction rates using CuHCF as the RR, which transports H^+ to pair sequentially with the HER and PSR reactions in two separate cells, denoted as Cell_{H_2} (left) and $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$ (right).

b, Chronopotentiometry curves of the electrodes in Cell_{H_2} (in logarithmic timescale) and $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$ (in linear timescale) when the ratio of the HER rate to PSR rate increases from 1:1 to 100:1 following the arrows. Here the PSR rate at all conditions was 10 C based on the capacity of the RR electrode, and the HER rate increased from 10 C to 100 C, 500 C and 1,000 C. The RR oxidation (blue) was paired with fast HER (orange), and the RR reduction (red) was paired with slow

PSR (green), both in 2 M H_2SO_4 solution. **c**, The average potential of the RR electrode (E_{RR} , top) and the corresponding average cell voltages (V_{cell} , bottom) of Cell_{H_2} and $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$ in 20-cycle ModES processes under varying ratios of the HER rate to PSR rate from 1:1, 10:1, 50:1 to 100:1. Blue and orange regions (in **c** and **d**) show different ModES processes. **d**, Efficiencies (Coulombic efficiency of the RR electrode, voltage efficiency of the ModES process and Faradaic efficiency of the persulfate production) and the concentrations of accumulated products during the ModES process under different operation conditions (corresponding to the same cycles in panel c). Conc, concentration.

respectively, in agreement with the rate-mismatched cycling results of the CuHCF electrode (Fig. 2f,g). The E_{RR} decreased by about 70 mV in $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$ and increased only about 170 mV in Cell_{H_2} with the increased HER rates (Fig. 3c). The V_{cell} did not change in $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$ and increased by nearly 350 mV in Cell_{H_2} (Fig. 3c). Although the HER rate increased dramatically, the average CE remained above 99.0%, the FE of PSR was ~70% and the VE of the ModES system was above 91.8% (Fig. 3d). These results confirm the tolerance and robustness of the decoupled ModES system operated at highly mismatched reaction rates. Consequently, this ModES system could effectively produce H_2 within 2 s and persulfate within 5 min, which makes it feasible to participate in different dynamic electricity markets.

Participation strategies in multiple electricity markets

We developed a computational modelling framework to simulate different electrochemical systems (coupled and decoupled), operating modes (flexible and inflexible power loads) and market participation strategies (Fig. 4a; more details in Supplementary Figs. 18–26 and Supplementary Note 1). This framework allows us to quantify the economic benefits of connecting a flexible, decoupled ModES system

to electricity markets operating at different timescales (DAM, RTM and FR) and at different geographical locations in the PJM (Pennsylvania, Jersey, Maryland) market and in the ERCOT (Electric Reliability Council of Texas) market. All systems and operating modes compared in this study produce the same total amount of chemicals. Figure 4b shows that the decoupled ModES system that operates under a flexible power load, particularly in PJM and Houston, TX (ERCOT), can reduce electricity costs by as much as 30–40% compared with the baseline system (a coupled system operating under an inflexible/constant power load). In areas with a strong incentive for market flexibility (high frequency of negative prices), such as the RTM in Panhandle, TX (ERCOT), operating under a flexible load can offset or even reverse electricity costs (generating revenue), highlighting the substantial economic benefits of providing demand flexibility to the power grid.

The decoupled ModES system has the ability to participate in the fast FR market to generate revenue (due to decoupling of reaction timescales), a flexibility feature that coupled systems cannot provide. To give more perspective on the benefits of participating in FR, we note that allowing the flexible ModES system to simultaneously participate in the DAM and in the FR market can generate up to 0.37 million $\text{US\$ yr}^{-1}$ in revenue from the FR market and reduce the total electricity cost by 65% (relative to a flexible ModES system that does not participate in

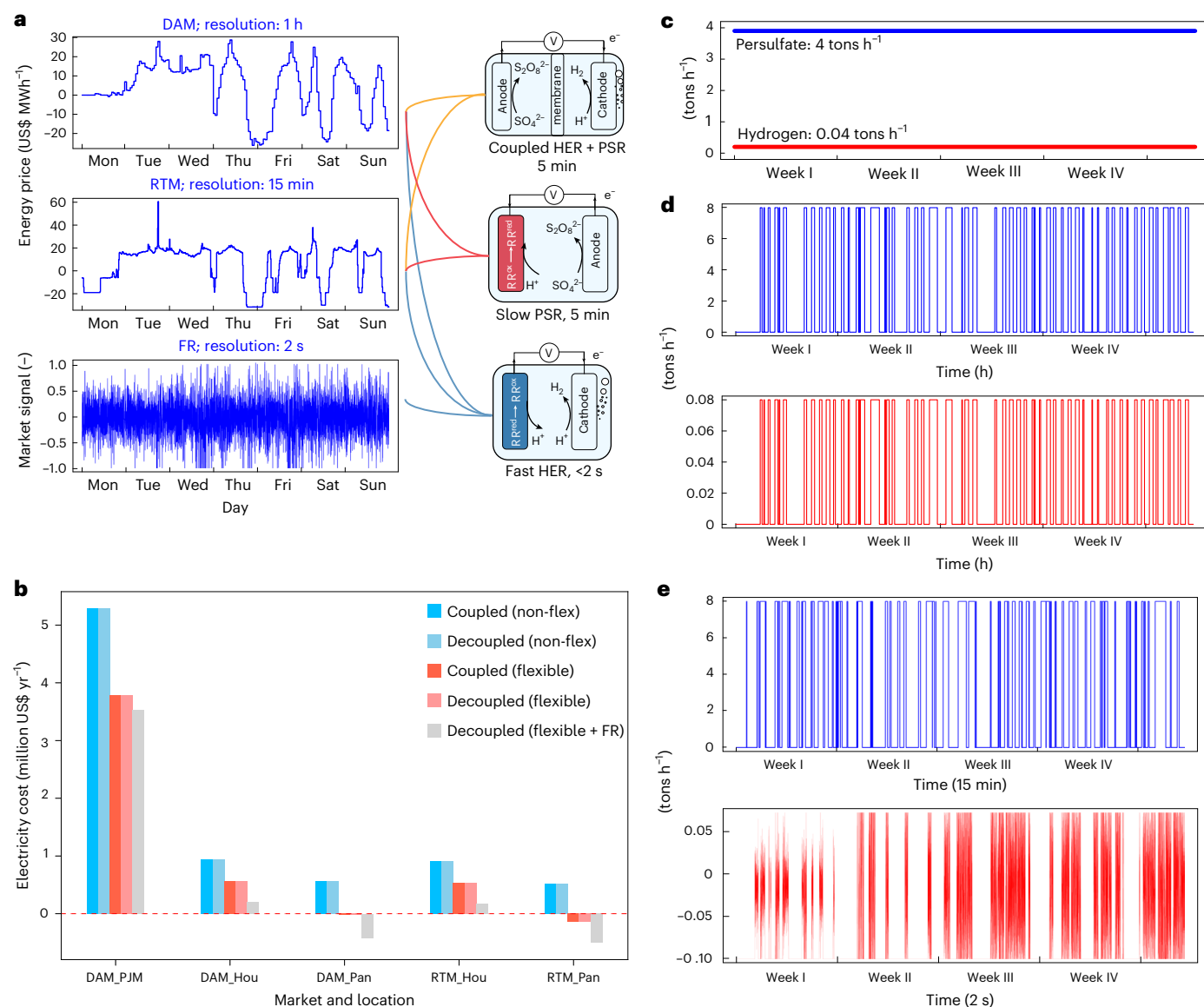


Fig. 4 | Market participation strategies and electricity cost reduction.

a, Different electricity markets under which coupled and decoupled electrochemical systems can participate. The conventional coupled system can only participate in DAM and RTM because its overall response time is limited by the persulfate production (PSR) and thus this system cannot respond to the fast FR market signals. The decoupled ModES system allows the fast H₂ production (HER) to track FR, DAM and RTM signals, whereas the slow PSR can participate in DAM and RTM. **b**, Quantification of the economic benefits of connecting the ModES system to different markets (DAM, RTM and FR) and at different locations, including PJM, Houston, TX (ERCOT), and Panhandle, TX (ERCOT). Overall, the use of flexible power loads leads to cost reductions and motivate the need to

track market signals. Moreover, decoupling enables cost reductions (in some cases leading to profit) due to participation in the FR market. **c**, Production schedule of a coupled system operated in constant mode under DAM for PJM. This operating mode is referenced as the ‘*Coupled (non-flex)*’ case in **b**. **d**, Production schedule for a coupled system showing how DAM market signals are followed to reduce costs. This operation mode corresponds to the ‘*Coupled (flexible)*’ case in **b**. **e**, Production schedule of the decoupled ModES system showing how decoupling enables the anode and cathode to follow markets of different time resolutions to reduce costs. This operation mode corresponds to the ‘*Decoupled (flexible + FR)*’ case in **b**.

the FR market). A substantial spatial variation in electricity costs was also observed. For instance, in the PJM market, the annual electricity costs for the flexible systems reach up to 3.5 million US\$ yr⁻¹, whereas in Panhandle, TX, the systems generate revenue. These results not only demonstrate that the flexibility of the decoupled ModES system can unveil additional revenue opportunities but also that such opportunities are strongly affected by electricity market conditions.

We also note that cost reductions require the synchronization of chemical production (anodic and cathodic) with market signals (full details in Supplementary Figs. 23–26). For instance, Fig. 4c shows the chemical production schedule of a coupled system operated

with constant power loads. In contrast, in the production schedule for a coupled system operated under flexible loads for the DAM in Houston, TX (Fig. 4d), the anode and cathode production schedules follow the slow dynamics of the DAM market. In the production schedule of the decoupled ModES system (Fig. 4e), it becomes apparent how persulfate production tracks the RTM market dynamics, whereas the hydrogen production tracks the fast FR market dynamics. This separation in production schedules is key for enabling cost reductions.

The analyses above show that strategically designing decoupled electrochemical synthesis can allow us to exploit the dynamic

electricity markets to generate more revenues using the demonstrated ModES system, which does not function as an energy storage system but a flexible load to the power grid that can produce valuable chemicals. Providing such demand flexibility generates more capital cost, therefore, we conduct a *preliminary* analysis of the capital expenditure for the ModES system and show that savings in electricity costs can offset the additional capital costs incurred by using the ModES system (Supplementary Note 2). We also note that our analysis assumes that the ModES system is a price taker and not an strategic bidder; capturing the interplay between bidding and market clearing procedures is an interesting analysis that requires more sophisticated optimization formulations (Supplementary Note 3).

Conclusions

We developed a modular electrochemical process with highly mismatched reaction rates that can participate in dynamic electricity markets to reduce the cost of electricity consumption for chemical production. Such a strategy is enabled by a fast proton battery material with excellent rate capability and electrochemical stability that delivers high Coulombic efficiency and capacity retention even under highly mismatched charging/discharging rates. The rate-mismatched ModES system achieves the co-production of hydrogen gas and persulfate with reaction rate ratio from 1:1 to 100:1 (when hydrogen production takes less than 2 s) and shows excellent Coulombic and energy efficiency ~100%. Furthermore, the developed computational framework shows that the ModES system operating under a flexible power load reduces electricity cost by 30–40% compared with the traditional coupled system operating under a constant power load. Such a decoupled ModES system can even generate revenue by participating in the FR market, which is impossible in the traditional coupled system. These results provide a conceptual strategy to help decarbonize the power grid by connecting flexible electrochemical synthesis to dynamic electricity markets and show how the decoupled electrosynthesis can open additional revenue opportunities.

Methods

Chemicals

All chemicals were purchased from Sigma Aldrich unless noted otherwise and used without further purification. Carbon black (Super P Conductive, 99.0+%) was purchased from Alfa Aesar. TUBALL BATT NMP 0.4% (a mixture of single-wall carbon nanotubes (SWCNT), 0.4 wt%; polyvinylidene fluoride, 2 wt%; *N*-methyl-2-pyrrolidone, >96.7 wt%) was purchased from OCSiAl. Titanium mesh was purchased from HeBei ChaoChuang Metal Mesh Co. Deionized nanopure water (18.2 = MΩ cm) from ThermoScientific Barnstead water purification systems was used for all experiments.

Synthesis of CuHCF

Copper hexacyanoferrate (CuHCF) was synthesized using a modified co-precipitation method³⁸. Typically, 40 ml of 0.2 M CuSO₄ solution was added dropwise into 40 ml of 0.1 M K₃Fe(CN)₆ solution under vigorous stirring at room temperature. After 6 h of reaction, the olive-green precipitate was centrifuged, rinsed with deionized water multiple times and dried in a vacuum oven at 60 °C overnight.

Materials characterization

Powder X-ray diffraction patterns of the CuHCF sample were collected using a Bruker D8 Advance X-ray diffractometer equipped with Cu-Kα radiation. The size and morphology of the samples were characterized using a scanning electron microscope (Zeiss SUPRA 55VP) equipped with an energy-dispersive X-ray spectroscopy detector. An inductively coupled plasma-optical emission spectrometer (Agilent 5110) was utilized to determine the compositions of K, Cu and Fe elements. Thermogravimetric analysis (TA Q500) was used to determine the water content in the CuHCF sample.

Fabrication and electrochemical tests of the RR electrodes

The RR electrodes were prepared via a conventional slurry-casting method using TUBALL BATT NMP 0.4%, Super P conductive carbon and the active materials. Typically, 70 wt% active materials and 18 wt% Super P carbon black were grounded for 30 min using pestle and mortar. Then the mixtures were added into TUBALL BATT NMP 0.4% that provided 2 wt% SWCNT and 10 wt% polyvinylidene fluoride for the electrode slurry. One ml NMP was added to the slurry for every 200 mg of active materials to control the viscosity. The slurry was stirred at 700 rpm overnight at room temperature and then cast onto titanium mesh current collectors (150 mesh, with a thickness of ~230 μm). The prepared electrodes were dried in a vacuum oven at 60 °C for 12 h to remove the residual solvent. The areal mass loading was 1 to ~2 mg cm⁻².

The electrochemical performance of the CuHCF RR electrode was characterized in 2 M H₂SO₄ solution in a three-electrode cell, with a Pt wire counter electrode and a saturated calomel electrode (SCE) as the reference electrode. Cyclic voltammetry (CV), and galvanostatic charge–discharge tests of the RR electrodes were recorded on a Bio-Logic VMP-3 multichannel potentiostat. The galvanostatic cycling was performed at 15 C rate for stability characterization and the rates of 1 C to 1,000 C for the kinetics characterization, where 1 C is defined as 65 mA g⁻¹ based on the theoretical capacity of as-synthesized CuHCF material. For studying the electrochemical behaviours in mismatched charging/discharging rates, the Coulombic efficiency (CE) and capacity retention (compared with that at 1 C) of the CuHCF electrode were evaluated when the charging rate was from 150 to 1,000 C, and the discharging rate was from 1.5 C to 750 C. The potentials of the RR electrodes were monitored using a Bio-Logic SP-200 potentiostat.

Fabrication of BDD electrode

The boron-doped diamond (BDD) electrode was prepared from a BDD/Si wafer (one side of Si and the other side of BDD, Lightning 25 DoSi wafer, Advanced Diamond Technologies Inc.). The wafer was cut to a 1 cm × 3 cm rectangular shape. After SiO₂ on the surface of the Si side was removed using 48 wt% HF, a piece of Cu foil was fixed by a piece of Cu tape and indium–gallium eutectic to make contact to the silicon water. Epoxy resin was used to seal the contact area and cover all edges and the back side (the Si side) and dried at room temperature overnight. The Cu foil was folded in a customized glass tube and sealed using epoxy to prevent solution contact, as shown in Supplementary Fig. 11.

Electrochemical characterization of the electrocatalytic electrodes

All electrocatalytic characterizations were performed at room temperature using a Bio-Logic SP-200 potentiostat or a Bio-Logic VMP-3 multichannel potentiostat. Hydrogen (H₂) and persulfate (S₂O₈²⁻) production reactions were first evaluated in 2 M H₂SO₄ solution in an H cell with a sintered glass filter disc (89057-758, ACE glass Inc.).

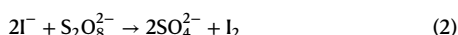
H₂ production. The hydrogen evolution reaction (HER) was evaluated in 2 M H₂SO₄ solution in the H cell using a Pt mesh electrode as the working electrode, a Pt mesh as the counter electrode and a SCE as the reference electrode. Linear scan voltammetry (LSV) curves were performed at a scan rate of 10 mV s⁻¹. Chronoamperometry (CA) was carried out at different potentials (–0.351, –0.301, –0.251 and –0.201 vs SCE). All potentials measured against SCE were converted to the standard hydrogen electrode (SHE) using $E_{\text{SHE}} = E_{\text{SCE}} + 0.241 \text{ V}$.

Persulfate production and detection. The persulfate (S₂O₈²⁻) production reaction (PSR) was performed in 2 M H₂SO₄ solution in an H cell with a glass filter, with a BDD electrode as the working electrode, a Pt wire as the counter electrode and a SCE reference electrode. The electrocatalytic performance of the BDD electrode was investigated via CV and LSV at 10 mV s⁻¹ and CA at different potentials (2.50, 2.60, 2.70 and

2.80 V vs SCE). All potentials measured against SCE were converted to the standard hydrogen electrode (SHE) using $E_{\text{SHE}} = E_{\text{SCE}} + 0.241 \text{ V}$.

To determine the Faradaic efficiency (FE) of the PSR, the concentration of the $\text{S}_2\text{O}_8^{2-}$ generated from CA was quantified by a spectrophotometric method based on iodometric titration. First, a $\text{Na}_2\text{S}_2\text{O}_8$ stock solution (20 mM) was prepared by dissolving $\text{Na}_2\text{S}_2\text{O}_8$ in deionized (DI) water. Then a series of standard solutions were prepared by mixing DI water, NaHCO_3 ($0.05 \text{ g } 10 \text{ mL}^{-1}$), KI ($1.0 \text{ g } 10 \text{ mL}^{-1}$) and various volumes of 20 mM sodium persulfate stock solution well and allowed to equilibrate for 15 min. The calibration curve was constructed by measuring the UV–vis spectroscopy of $\text{Na}_2\text{S}_2\text{O}_8$ solutions using a Cary 50 SCAN spectrometer with different concentrations. The wavelength used for the spectrophotometric measurement is 352 nm.

The sample solution containing $\text{S}_2\text{O}_8^{2-}$ with a proper volume was added into 10 mL NaHCO_3/KI chromogenic solution that contained 0.05 g NaHCO_3 and 1.0 g KI and measured by UV–vis spectroscopy. The concentration of $\text{S}_2\text{O}_8^{2-}$ could be determined by the following equations:



$$[\text{S}_2\text{O}_8^{2-}] (\text{mM}) = \frac{(V_1 + \Delta V) \times [\text{S}_2\text{O}_8^{2-}]_{\text{detected}}}{\Delta V} \quad (3)$$

where V_1 , ΔV and $[\text{S}_2\text{O}_8^{2-}]_{\text{detected}}$ are the original volume of NaHCO_3/KI solution, the volume of added $\text{S}_2\text{O}_8^{2-}$ sample solution and the detected concentration of $\text{S}_2\text{O}_8^{2-}$ in the analysed solution after adding $\text{S}_2\text{O}_8^{2-}$ sample, respectively.

The FE of $\text{S}_2\text{O}_8^{2-}$ production reaction was then calculated based on the detected concentration of $\text{S}_2\text{O}_8^{2-}$ and input charge using the following equation:

$$\text{FE} (\%) = \frac{Q_{\text{for } \text{S}_2\text{O}_8^{2-} \text{ production}}}{Q_{\text{input}}} \times 100 = \frac{[\text{S}_2\text{O}_8^{2-}] \times V \times 2 \times 96,485}{Q_{\text{input}}} \times 100 \quad (4)$$

where V , $[\text{S}_2\text{O}_8^{2-}]$ and Q_{input} are the volume of the solution, the concentration of produced $\text{S}_2\text{O}_8^{2-}$ and the input charge during the electrosynthesis, respectively.

Demonstration of highly rate-mismatched ModES process

Two electrochemical cells, the H_2 cell (Cell_{H_2}), and the $\text{S}_2\text{O}_8^{2-}$ cell ($\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$), which both contained 35 mL 2 M H_2SO_4 solution, were used. A CuHCF RR electrode with a mass loading of 7.86 mg (for the HER rate/PSR rate of 50:1 or 100:1) or 20.8 mg (for the HER rate/PSR rate of 1:1 or 10:1) was used as the counter electrodes in different ModES scenarios. The total capacity of used CuHCF electrodes and the number of moles of H_2 /persulfate produced at different mismatched rates in one ModES reaction cycle are summarized in Supplementary Table 3. In Cell_{H_2} , a Pt mesh and a SCE were used as the working and reference electrodes. In $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$, a BDD electrode ($-1 \times 2 \text{ cm}^2$) and a SCE were used as the working and reference electrodes, respectively.

Typically, in Cell_{H_2} , the RR oxidation from RR^{red} to RR^{ox} released H^+ until 1.10 V vs. SCE (1.341 V vs SHE), and H_2 was generated at the cathode. Then, the RR^{ox} electrode was taken out of the Cell_{H_2} , washed with 2 M H_2SO_4 solution to remove the residual electrolyte and moved to $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$. In $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$ the RR^{ox} was reduced to RR^{red} and consumed H^+ until 0.50 V vs SCE (0.741 V vs SHE) and $\text{S}_2\text{O}_8^{2-}$ was produced at the BDD electrode. For highly rate-mismatched ModES processes, the ratios of the HER rate to the PSR rate were 1:1, 10:1, 50:1 and 100:1, where the PSR rate was kept at 10 C based on the capacity of the CuHCF electrode used with the corresponding HER rates of 10 C, 100 C, 500 C and 1,000 C. The area of the BDD electrode in 2 M H_2SO_4 solution was modified based on the applied current to maintain the similar applied potential for PSR. The RR electrode's potential was monitored to calculate the total

cell voltage (V_{cell}), that is, the potential difference between the RR electrode and the Pt or BDD electrode. During these ModES cycles, the capacity of the RR electrode was controlled by the potential and capacity restrictions to ensure the RR electrode was operated within the desired potential windows.

The concentrations of the produced $\text{S}_2\text{O}_8^{2-}$ was measured following the procedures described above. On the basis of the product concentrations, the Faradaic efficiencies were calculated according to the following equation:

$$\begin{aligned} \text{FE} (\%)_{\text{S}_2\text{O}_8^{2-}} &= \frac{Q_{\text{for } \text{S}_2\text{O}_8^{2-} \text{ production}}}{Q_{\text{passing,RR}}} \times 100 \\ &= \frac{[\text{S}_2\text{O}_8^{2-}] \times V \times 2 \times 96,485}{Q_{\text{passing,RR}}} \times 100 \end{aligned} \quad (5)$$

where the $Q_{\text{passing,RR}}$ is the charge passing through the RR electrode in the $\text{Cell}_{\text{S}_2\text{O}_8^{2-}}$.

Data availability

The frequency regulation market data was obtained from the PJM Interconnection LLC Independent System Operator. Frequency regulation price data is public, but the frequency regulation signal data is not publicly available. Source data for figures that do not use the frequency regulation market data are provided with the manuscript. Source data are provided with this paper.

Code availability

The code and necessary data for the computational framework are provided as a Supplementary Code file with the manuscript.

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Author contributions

R.W., H.S., V.M.Z. and S.J. conceived the idea and designed the experiments. R.W. carried out materials synthesis, materials characterization, electrochemical measurements and electrochemical production. J.M. and V.M.Z. conducted numerical analysis. V.M.Z. and S.J. supervised the project. R.W., J.M., V.M.Z. and S.J. wrote the manuscript, and all authors commented on it.

Competing interests

The authors declare no competing interests.

Additional information

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